Aaliyan Muhammad

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Education

The University of Western Ontario — London, Ontario
Bachelor of Science · Computer Science — *Expected 2027*

Experience

Software Developer Intern — May 2026 – Aug 2026
Guidewire Software — *Mississauga, ON*

- Resolved and shipped **50+ production defects** across **PMIC's enterprise insurance platform** using **React**, **HTML**, **CSS/SASS**, **TypeScript**, and the **Jutro design system** within a scalable micro-frontend architecture.
- Implemented **user stories** for the Agent Experience application, including **Whatfix integration** and **Dockerized test deployments**, collaborating with cross-functional stakeholders to translate requirements into production-ready features.
- Managed **15+ application and micro-frontend deployments** across **Agile workflows** using **Bitbucket**, **TeamCity**, and **Azure DevOps** to support **CI/CD** and reliable release delivery.
- Designed and presented a **AI-assisted testing workflow** that automated repetitive defect setup, **saving ∼10 minutes per test** and accelerating validation across the engineering team.

Programming Tutor — Nov 2024 – Oct 2025
EduXora — *London, ON*

- Delivered customized **Java** and **C/C#** instruction to over **100** first-year students through structured lessons, guided practice, targeted examples, and feedback increasing average test scores by **20%** and coursework grades by **25%**.
- Led interactive problem-solving sessions focused on reinforcing core concepts and applied reasoning skills, encouraging participation and discussion, resulting in a **30% increase in student engagement** and stronger confidence.

Projects

Career Compass | *React, TypeScript, Spring Boot, Python, PostgreSQL, AWS, Docker* | *Live* — Sept 2025 – Present

- Engineered a **full-stack career platform** using **React**, **TypeScript**, **Spring Boot**, **Python**, and **PostgreSQL**, enabling authenticated tracking of up to **1,000** applications per workspace with persistent data and document management.
- Developed a **RAG-based Career Copilot** using **OpenAI** embeddings and **PostgreSQL/pgvector**, implementing document chunking and vector retrieval to generate responses grounded in resumes, job descriptions, and project notes.
- Architected and containerized separate web, **Spring Boot API**, and **Python AI** services with **Docker**, integrating **REST APIs**, server-side validation, PostgreSQL persistence, and secure service-to-service communication.
- Deployed production infrastructure to **AWS** with **Terraform** using ECS/Fargate, RDS, S3, Cognito, ECR, ALB/ACM, Secrets Manager, and **CloudWatch**, delivering an **HTTPS-secured application** at careercompass.aaliyanm.dev.

audioDecoded | *Python, FastAPI, PostgreSQL, Librosa, NumPy, JavaScript, DSP* | *Live* — Feb 2026 – Sept 2026

- Built a **full-stack audio analysis web app** using **Python**, **FastAPI**, and **Librosa** to detect BPM and musical key from uploaded or recorded audio via real-time **DSP pipelines**, achieving **∼98% accuracy** across benchmark test tracks.
- Developed a responsive single-page interface with **HTML**, **CSS**, and **JavaScript** supporting drag-and-drop uploads and live microphone input, **processing audio files in under 2 seconds** with real-time result visualization.
- Implemented **DSP algorithms** using **Librosa**, **NumPy**, and **SciPy**, leveraging harmonic–percussive separation, chroma features, and onset strength analysis to evaluate **150+ audio samples** and improve tempo estimation reliability.
- Integrated **PostgreSQL** to persist users, audio uploads, and analysis results, enabling analysis history and future expansion while minimizing unnecessary database writes.

Modular Study Optimization System | *C++, Qt, OOD, Modular Architecture* | *GitHub* — June 2025 – Dec 2025

- Built a **cross-platform study optimization desktop app** using **C++** and **Qt** to generate adaptive schedules from user inputs, implementing **modular OOD architecture** to process **50+** constraints and manage **100+ tasks.**

Technical Skills

Languages: Java, Python, C, C#, C++, Assembly, JavaScript, TypeScript, HTML/CSS, SQL
Frameworks & Libraries: React, Spring Boot, Node.js, Express, FastAPI, Tailwind CSS, Framer Motion, LangChain
APIs & Databases: PostgreSQL, RESTful APIs, OpenAI API
Developer Tools: Git, GitHub, Docker, AWS, Terraform, Azure DevOps, CI/CD, SDLC, Microsoft Azure, Linux/Unix
Systems: Windows Server 2022 installation & configuration, Windows 11 & Hyper-V management
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Intro to A.I, Software Architecture